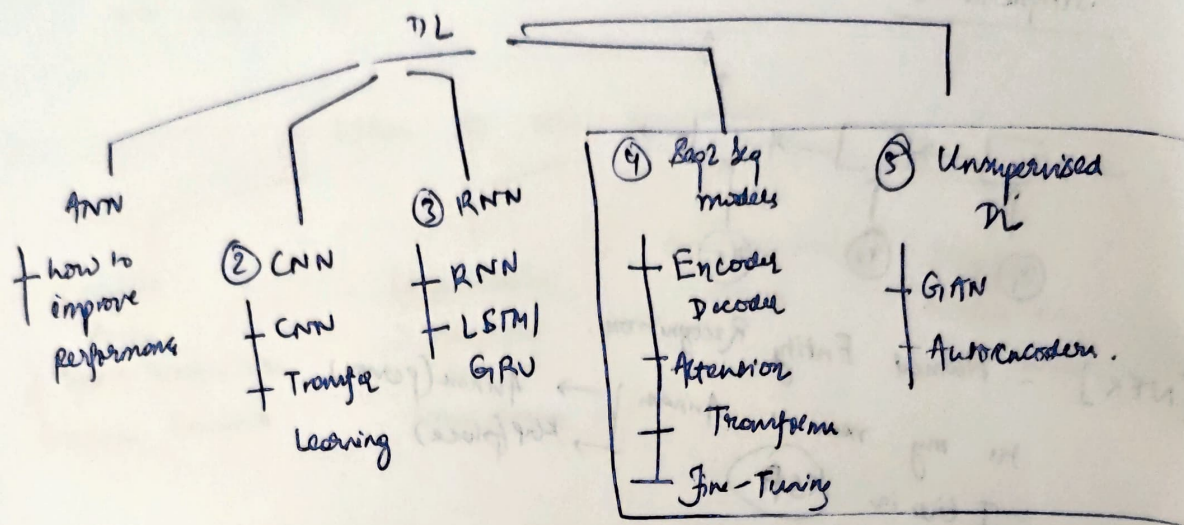


Day 64

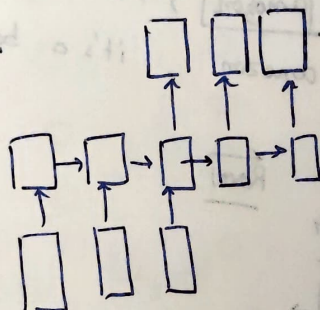
Lec -67 | History Time

LLMs



Seq2Seq Models

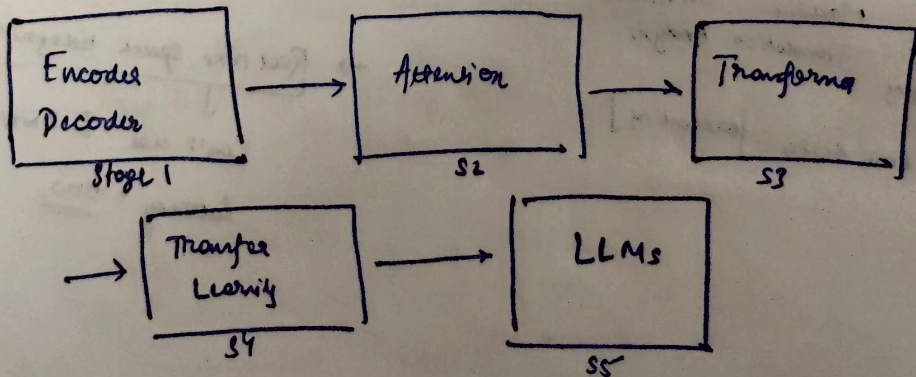
1. Asynchronous Models.



Apps.

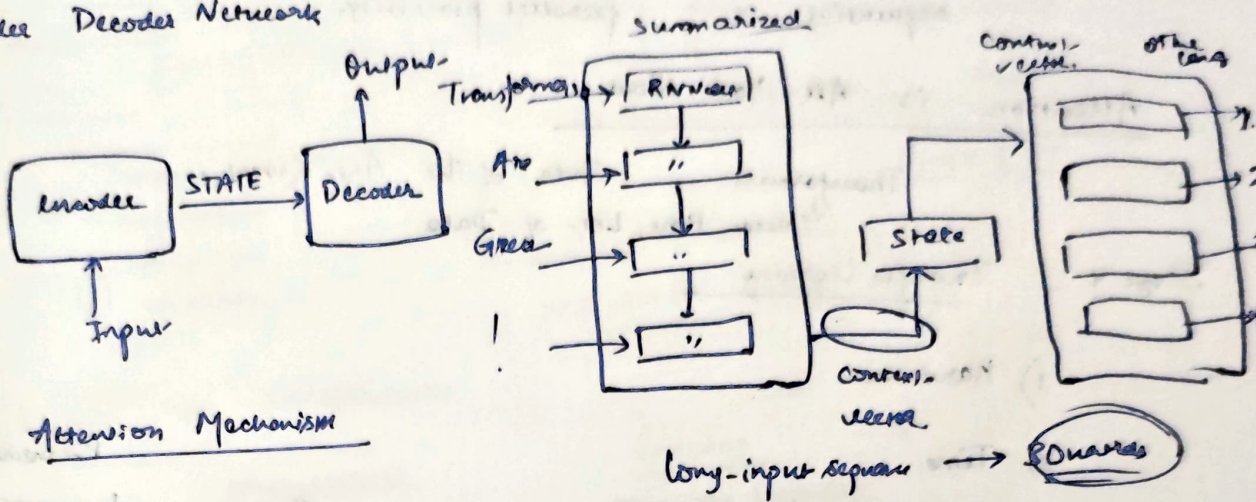
- 1) text summarization → [Seq] → [Seq]
- 2) Question Ans. → [Seq] → [Seq]
- 3) Chatbot → input (text) → output (text)
- 4) Speech to text

History of Seq2Seq Models

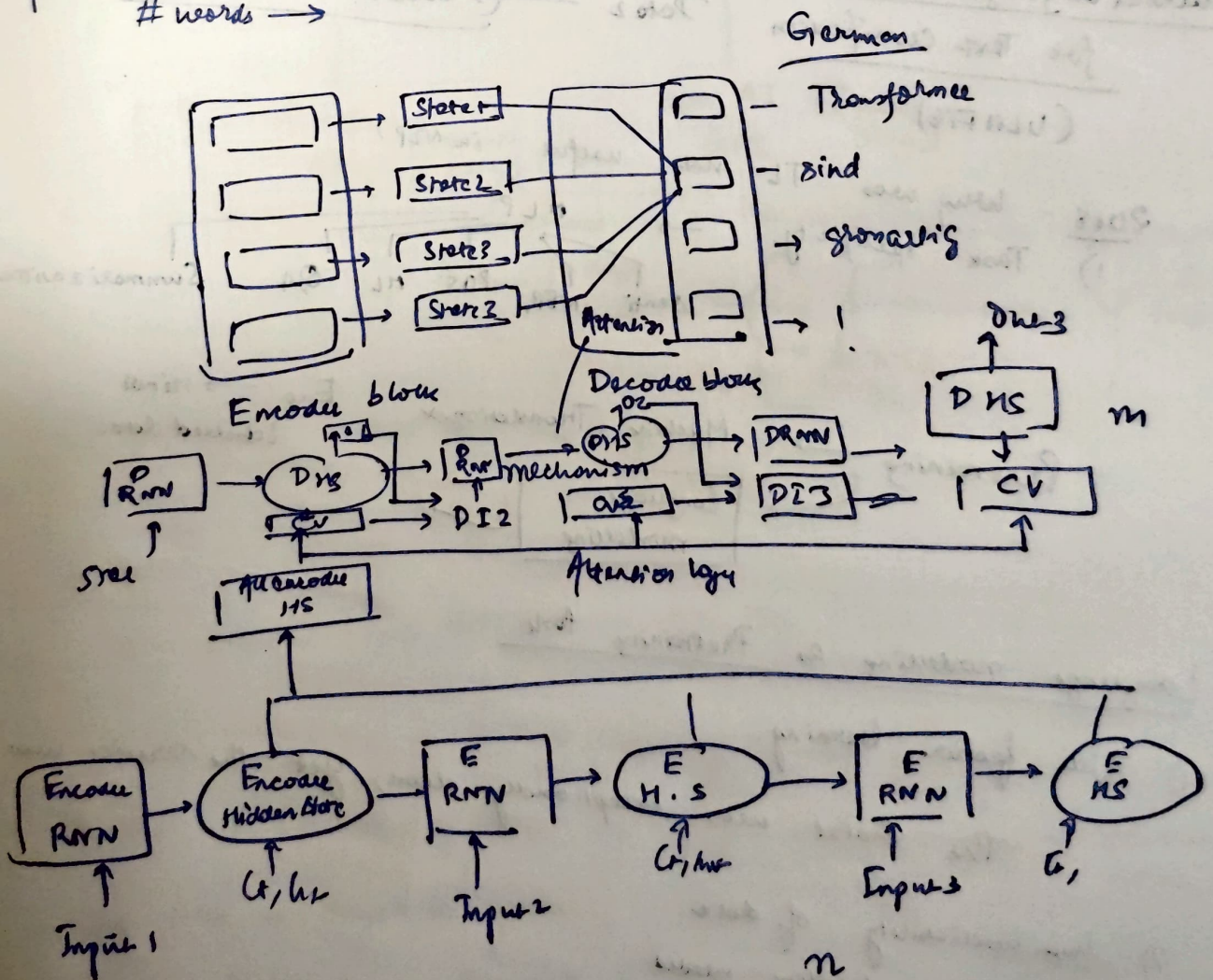
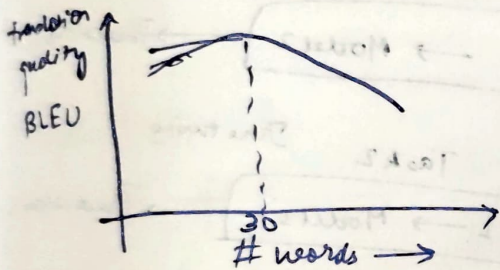


Ilya Sukker (Co-founder OPEN AI)

Encoder Decoder Networks



Stage-2 Attention Mechanism



$\boxed{n \times m} \rightarrow \text{quadrilateral complexity.}$

Stage 3 - Transformers

sequential → parallel processing

Attention is All You Need

Transformers - State of the Art - Result.
↓ Needs time, Lot of Data

Stage 4 - Transfer Learning

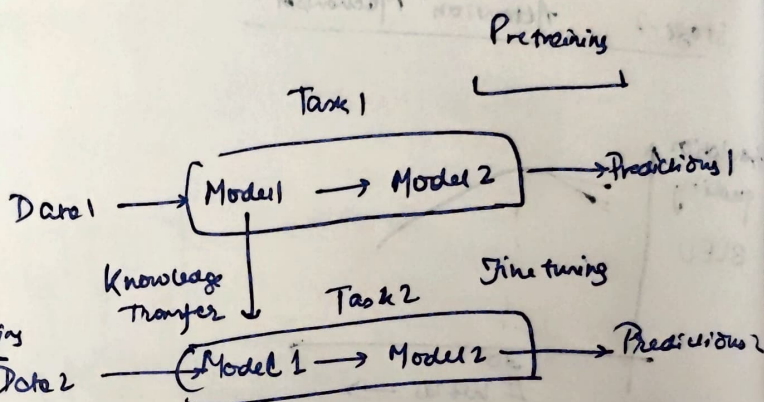
1) Hardware

2) Time

3) Data → a lot

Transfer Learning

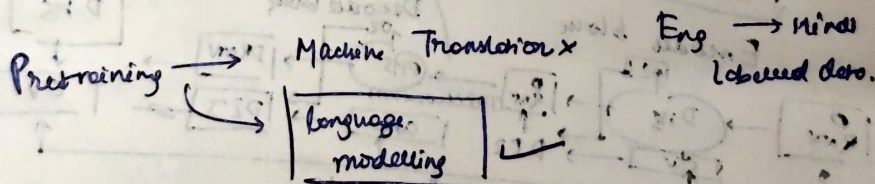
Universal Language Model Fine-tuning for Text Classification (ULMFit)



2018 - Why was TL not useful in NLP?

1) Task specificity.

NLP tasks: Sentiment, NER, POS, ML, QA, Summarization



Language modelling as Pretraining task

1) Rich feature learning.

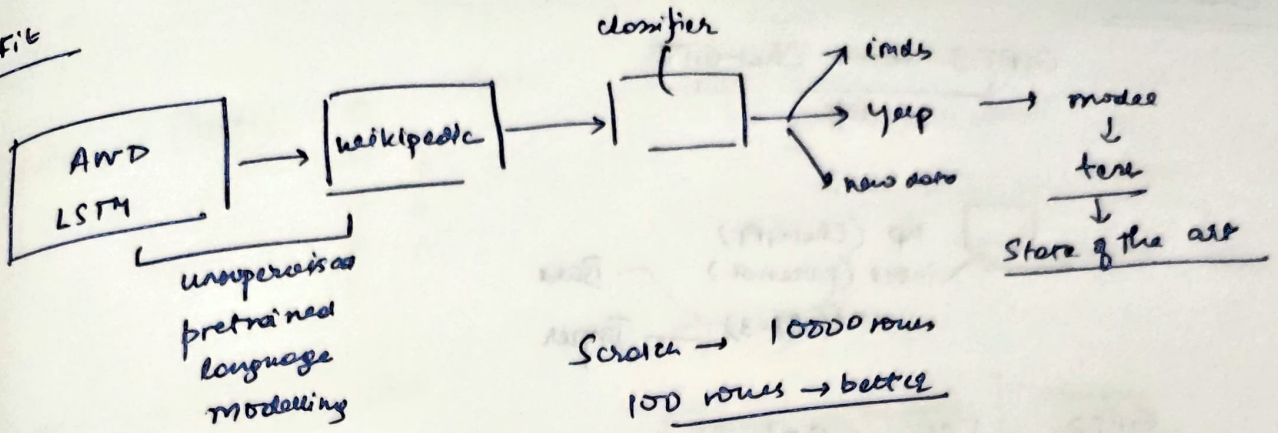
The hotel was exceptionally clean, yet the service was bas pathetic

2) huge availability of data.

- No labelling needed

- Unsupervised task.

ULMFiT

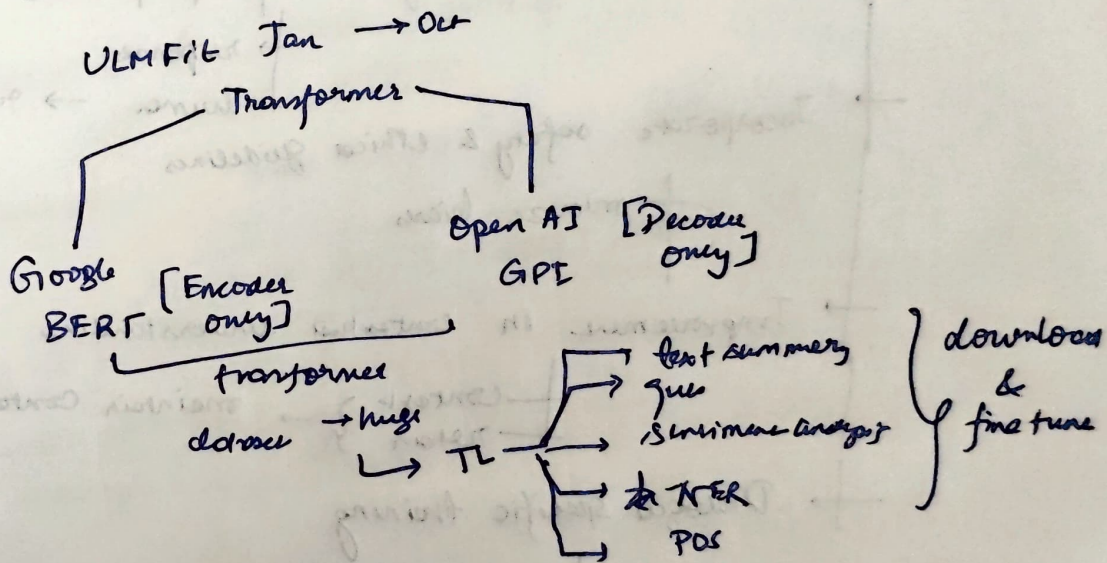


Didn't use transformers.

architecture
Transformers

training
Transfer learning.

Stage-5 LLM
2018



OPEN AI → GPT-2 → GPT-3
Large Language Model

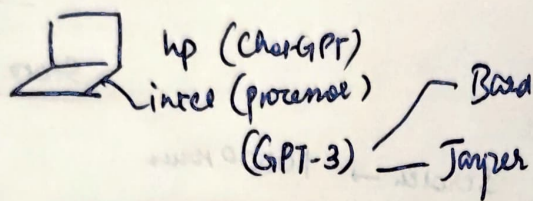
Qualities of LLMs

- 1) Data → billions → GPT3 → 45 TB
books, websites, internet
diversity → bias
- 2) Hardware → Cluster of GPU → GPT3 → Supercomputer → 1000s NVIDIA GPUs
- 3) Training time → days to weeks.
- 4) Cost → hardware + electricity + infra + experts.
- 5) Energy consumption → month small town

+ company
+ govt
+ institutes
+ Energy.

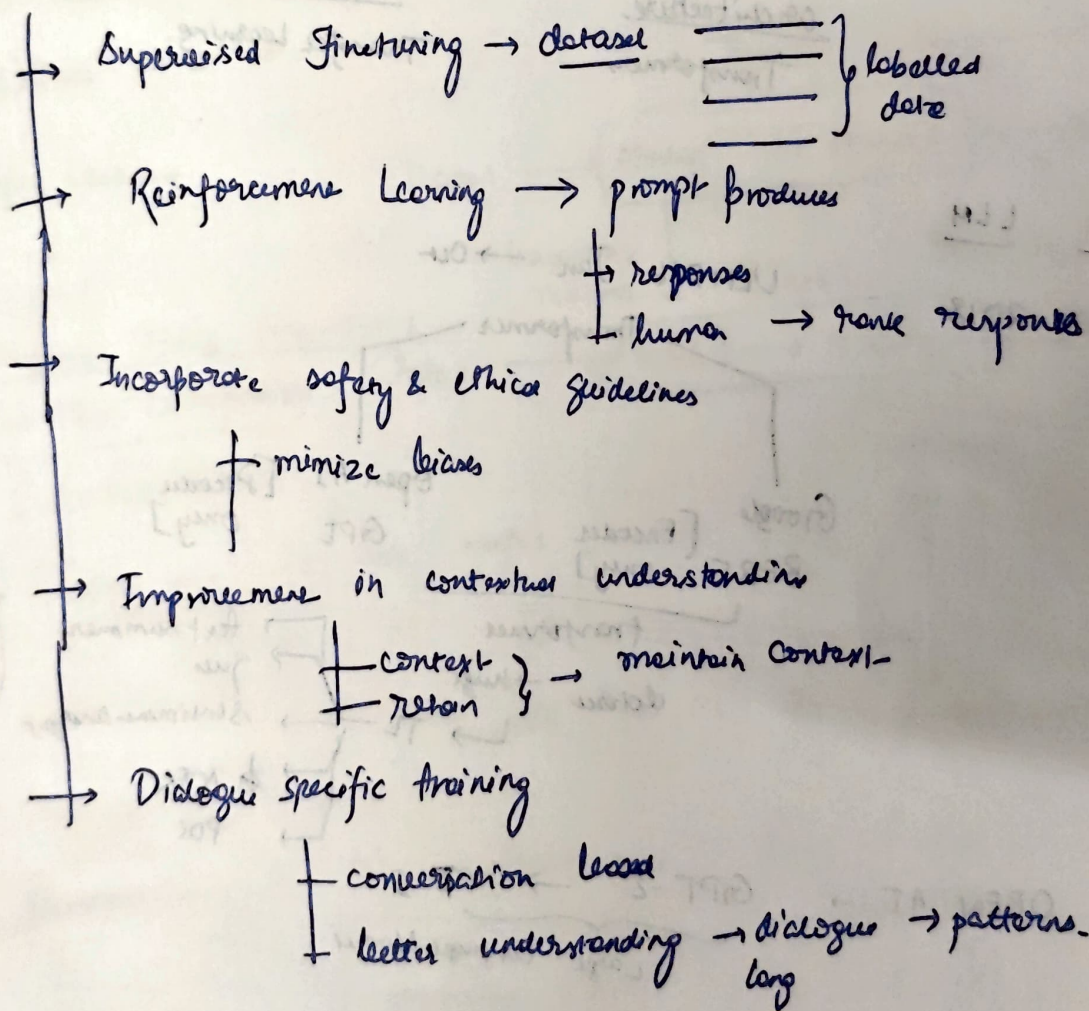
Chat GPT

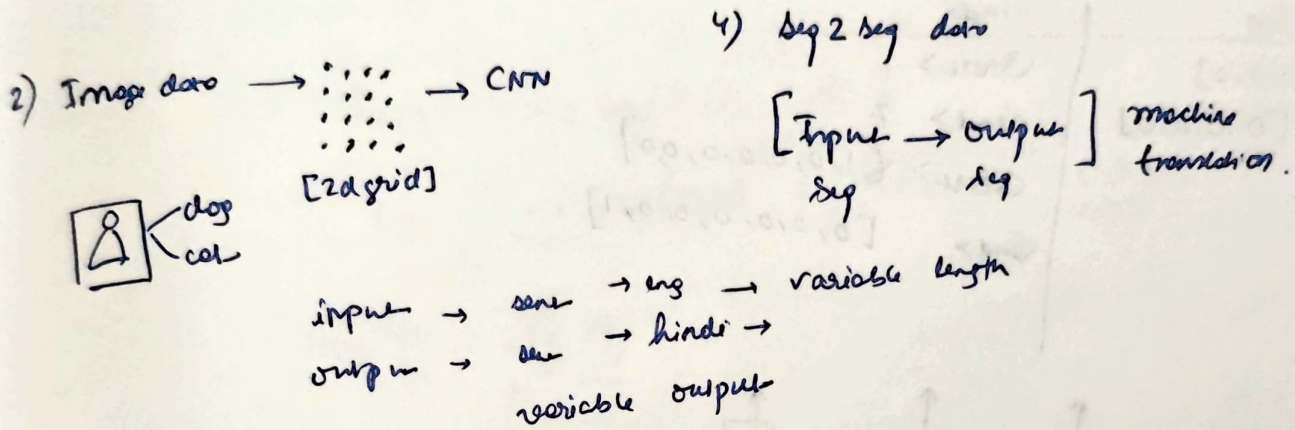
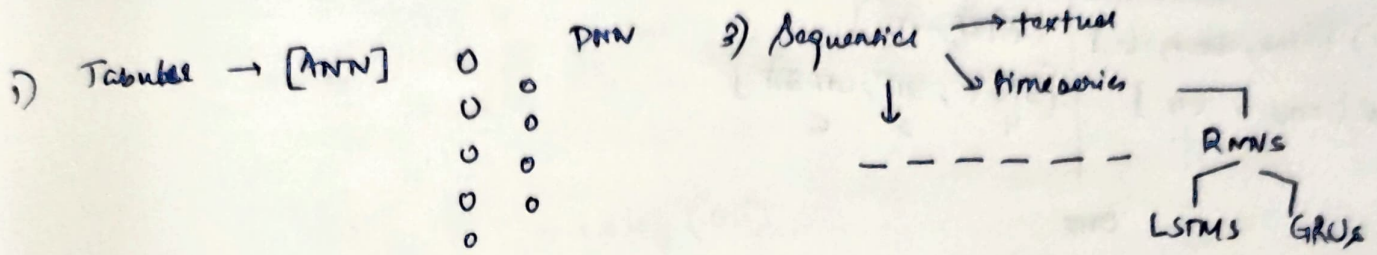
GPT-3 ↔ Chat GPT
diff.



GPT-3 — [Chat-GPT]

1) RLHF → Reinforcement Learning from Human Feedback





Sequence to Sequence Learning with Neural Networks

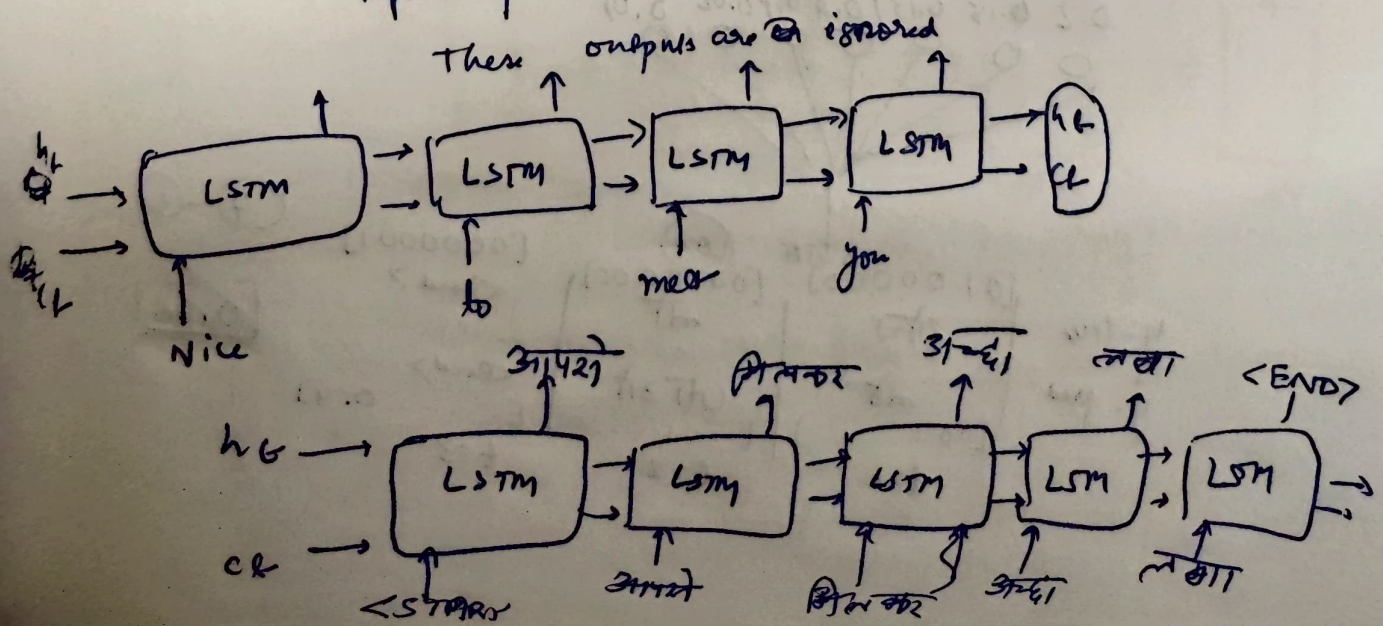
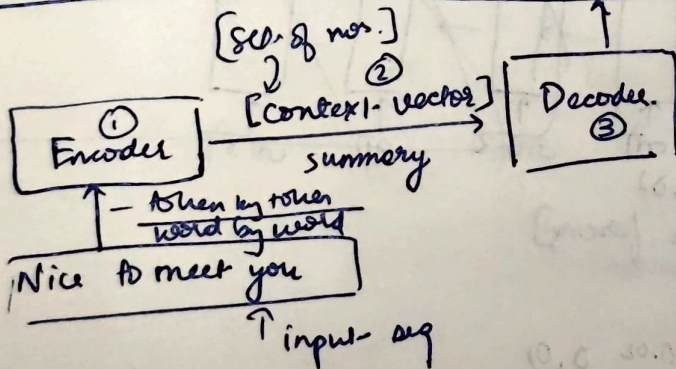
Ilya Sutskever

Oriol Vinyals

Quoc V. Le

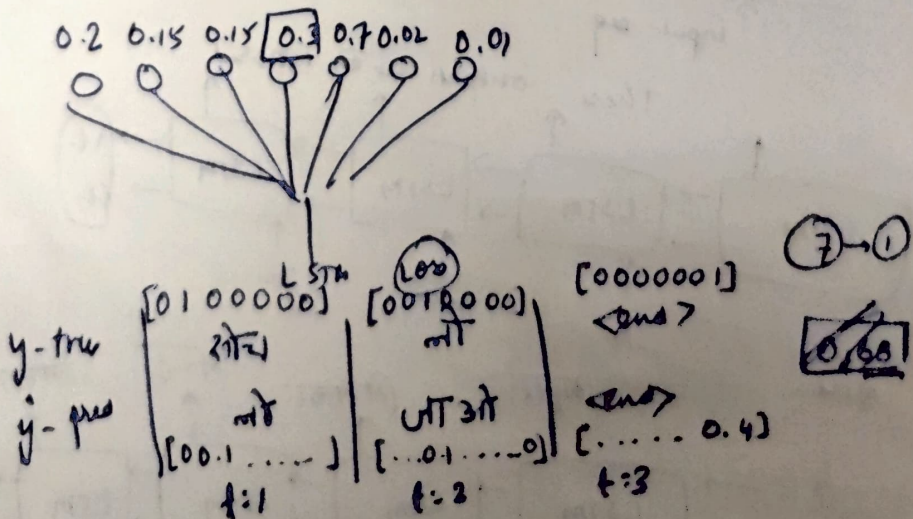
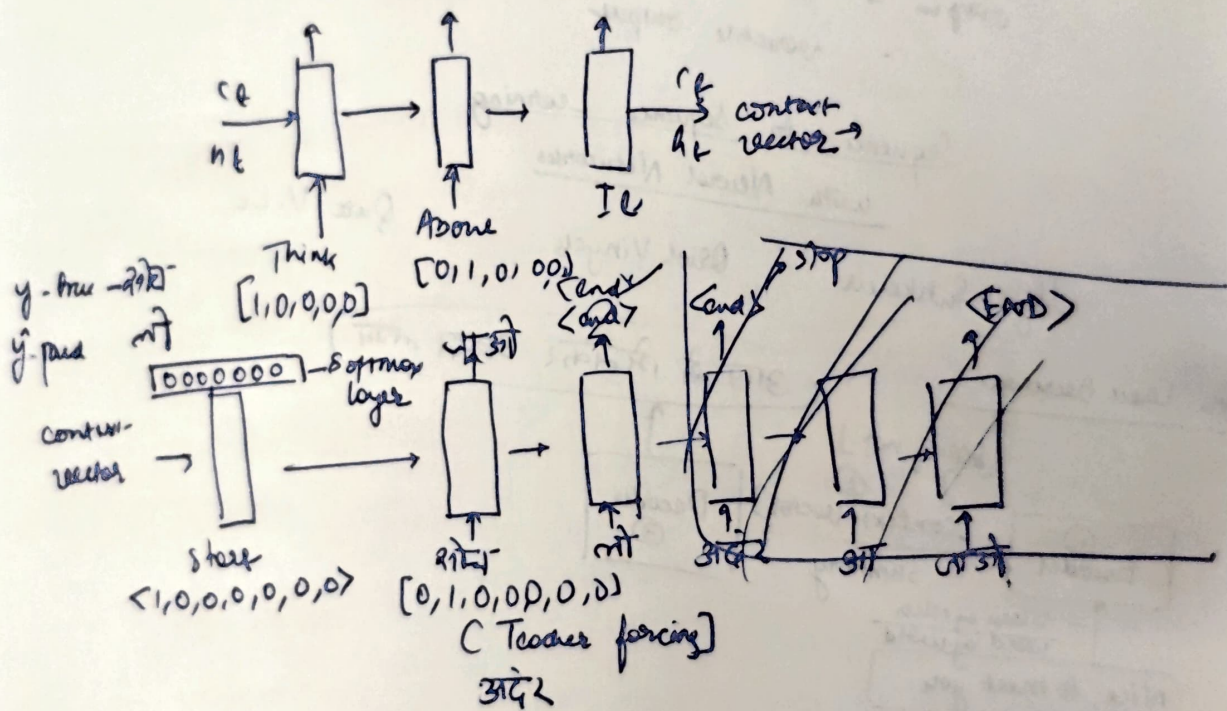
High Level Overview

आप से मिलकर अर्द्धा लगी



Eng	hindi
1) [think, about, it]	[सोचो, मो]
2) [come, in]	[आदर, आ, जाओ]

Eng	hindi
[1, 0, 0, 0, 0]	<start> 1
[0, 1, 0, 0, 0]	<end> 7
	<start> [1, 0, 0, 0, 0, 0]
	<end> [0, 0, 0, 0, 0, 1]



Loss $\rightarrow$ categorical cross entropy

$$L = - \sum_{i=1}^7 y_i^{\text{true}} \log(y_i^{\text{pred}})$$

$$L_{t=1} = -1 \times \log(0.1)$$

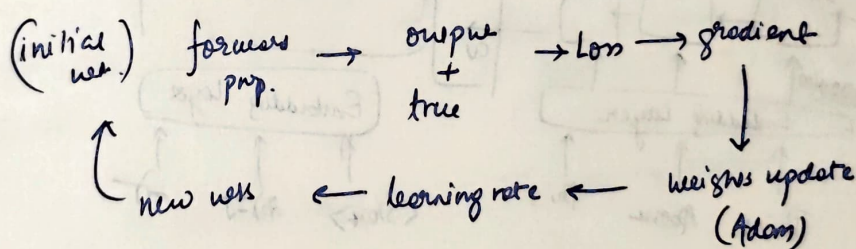
$$= 1$$

$$L_{t=2} = -1 \log(0.1) = 1$$

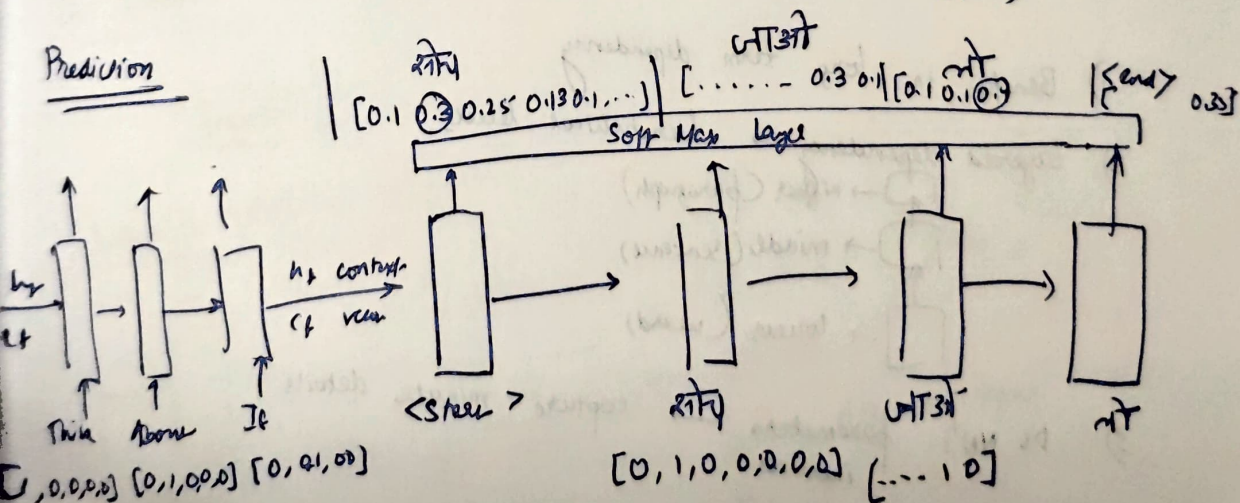
$$L_3 = -1 \times \log(0.4) = -1 \times 0.39 = \underline{\underline{0.39}}$$

$$\text{avg loss} = \frac{1}{7} (2.39)$$

- 1) ~~for~~ forward prop
- 2) Loss
- 3) Backpropagation
 - $\rightarrow$ gradient calculate
 - $\rightarrow$ update parameters.



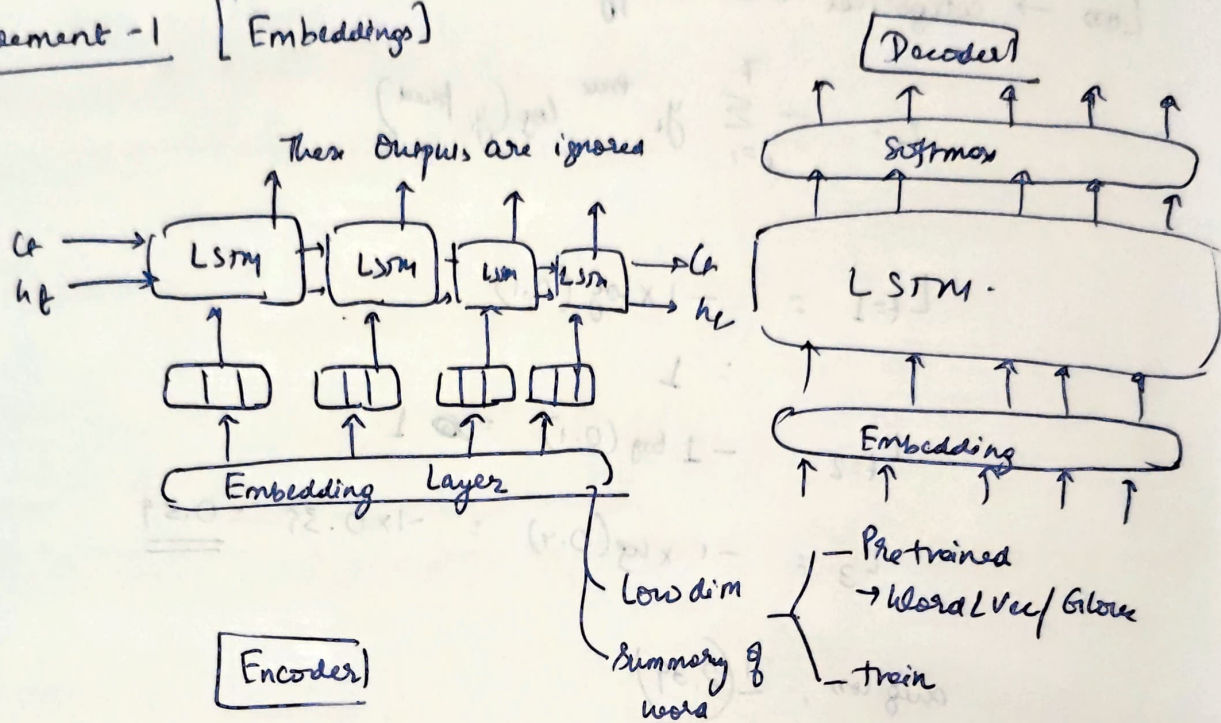
Prediction



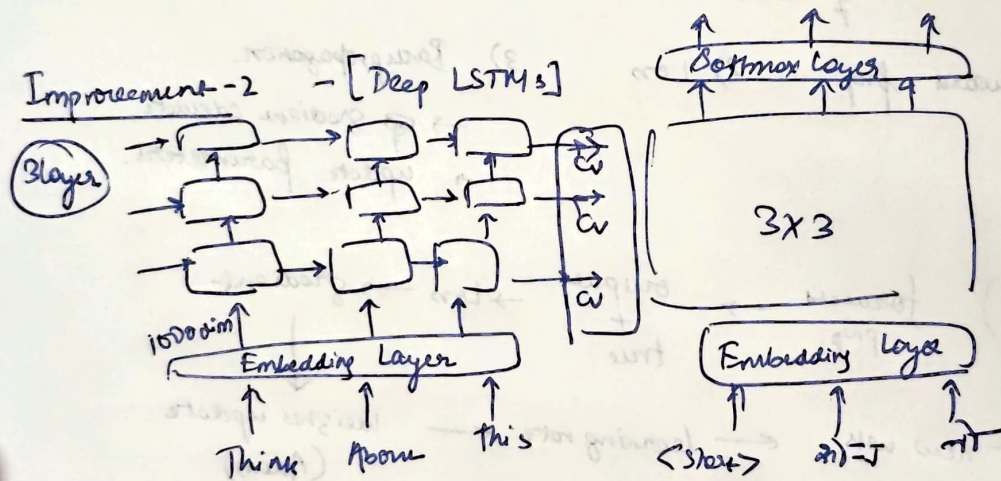
input $\rightarrow$ [Things about It]

output $\rightarrow$ સોચ જાણ નો

Improvement - 1 [Embeddings]



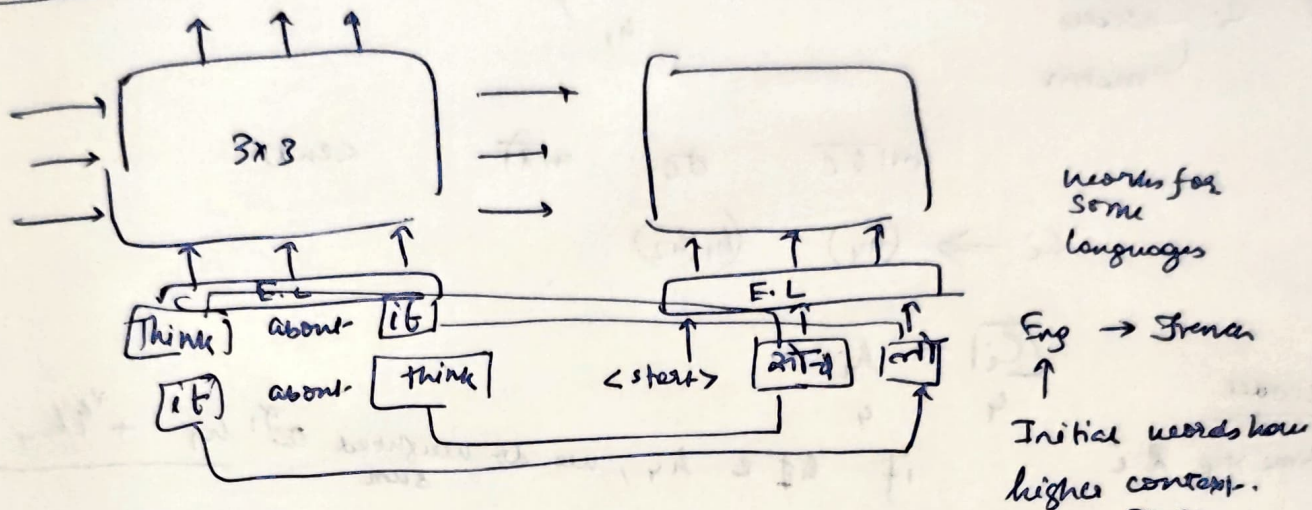
Improvement - 2 - [Deep LSTMs]



- 1) Benefit in long term dependency
- 2) Layered dependency $\rightarrow$ hierarchical levels.
 - $\rightarrow$ higher (paragraph)
 - $\rightarrow$ middle (sentence)
 - $\rightarrow$ lower (word)
- 3) DL NN's parameters $\rightarrow$ capture minute details increases

Original paper $\rightarrow$ 4 layer Deep LSTMs

Improvement 3 - Reversing the Input



Performance → BLEU Score - The model achieved a BLEU score of 34.81, surpassing the base line Statistical Machine Translation (SMT) system's score of 33.30 on the same dataset, marking a significant achievement.

Lec-69 Attention Mechanism

The why?

> 25 words
Problem

① Encoder
Summary-load

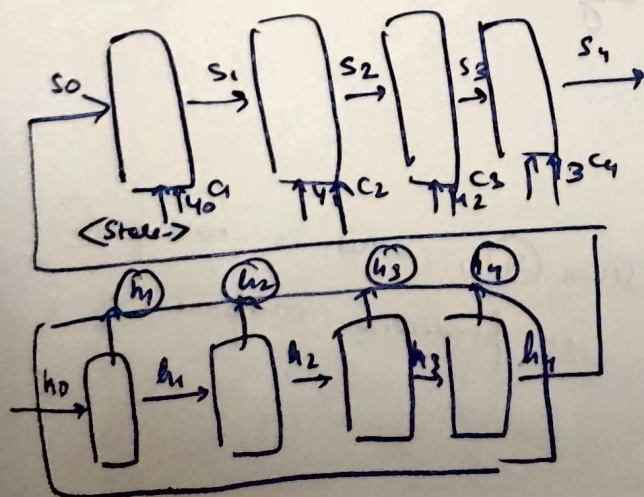
Decoder

② We need specific set of words.
Because of static representation
Decoder has problem in printing output

The solution — Attention Mechanism

Important

What is Attention Mechanism?



$i=2$

$[y_1, s_1]$ 2 pieces

Attention

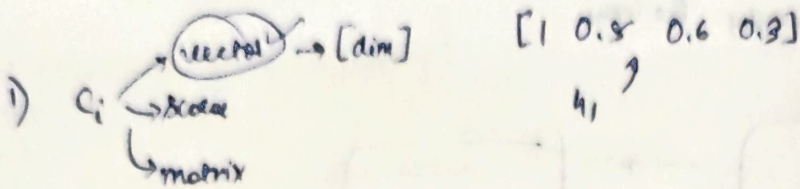
$[c_i]$ - Attention input

$i=2$

$[y_1, s_1, c_2]$

①

$[y_{i-1}, s_{i-1}, c_i]$



मॉडल स्टेट क्वैरी <end>

$$C_i \Rightarrow (h_4) \quad (h_1, h_2)$$

$$\begin{matrix} [C_i] & \rightarrow & h_i \\ 4 & & 4 \end{matrix}$$

decoder

time step i

if h_1 & h_4 , we do weighted sum $\alpha_1 h_1 + \alpha_4 h_4$

$$C_1 = \alpha_1 h_1 + \alpha_2 h_2 + \alpha_3 h_3 + \alpha_4 h_4$$

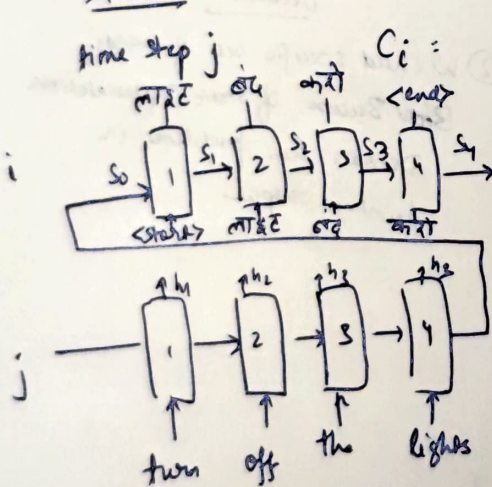
if $\alpha_4 \uparrow \Rightarrow h_4$ will be dominant

$$C_2 = \alpha_1 h_1 + \alpha_2 h_2 + \alpha_3 h_3 + \alpha_4 h_4$$

encoder

time step j

$$C_i = \sum \alpha_{ij} h_j$$



$$C_1 = \alpha_{11} h_1 + \alpha_{12} h_2 + \alpha_{13} h_3 + \alpha_{14} h_4$$

$\alpha_{21} \rightarrow$ alignment / similarity score

DEncoder $i=2 \rightarrow$ Output

Encoder $j=1$ (Rde)

$$\alpha_{21} \rightarrow f(h_1, s_1)$$

h_1 [s] $\rightarrow$ previous hidden state of decoder

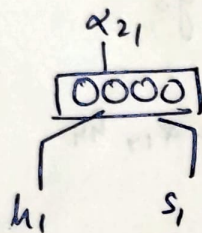
Given (s_1) , what is role of (h_1) in predicting output of $i=2$

$$\alpha_{ij} \rightarrow f(h_j, s_{i-1})$$

$$\alpha_{23} \rightarrow f(h_3, s_1)$$

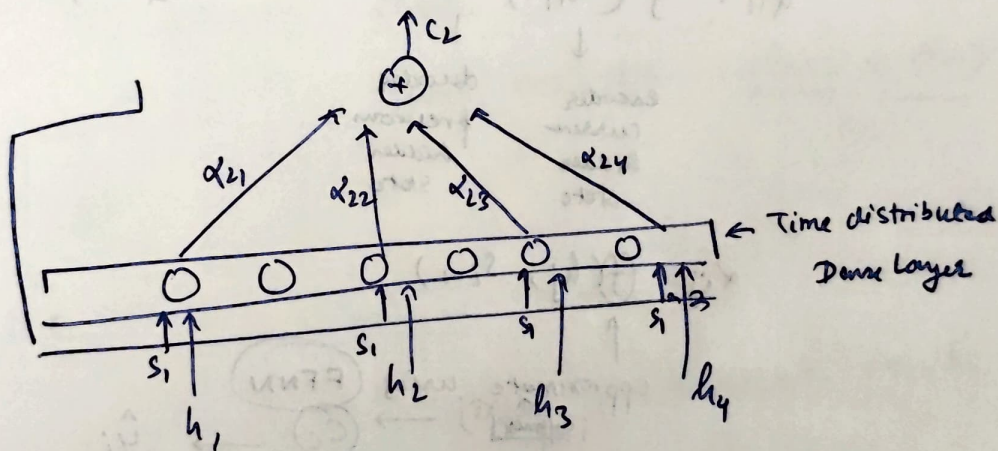
what function should we use? $\rightarrow$ ANN

ANN - universal function approximation

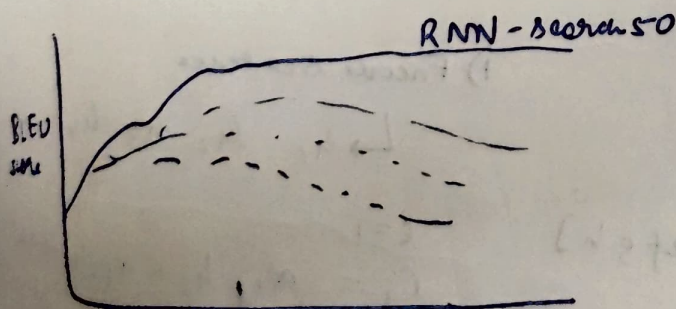


α_{ij}

$$C_2 = \alpha_{21} h_1 + \alpha_{22} h_2 + \alpha_{23} h_3 + \alpha_{24} h_4$$



Hence, $i=2$ $[C_2, s_1, y_2]$



i	j	turn off the lights			
		1	2	3	4
1	MISC	α_{11}	α_{12}	α_{13}	α_{14}
2	बंद	α_{21}	α_{22}	α_{23}	α_{24}
3	करो	α_{31}	α_{32}	α_{33}	α_{34}
4	करा	α_{41}	α_{42}	α_{43}	α_{44}

This Grid shows It works

* In original paper, Bidirectional LSTMs were used in Encoder.

Lec-70 Bahdanau Vs Luong Attention

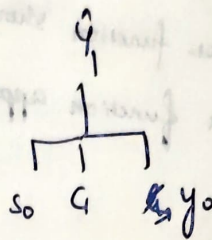
[NMT] → Neural Machine Translation

Decoder i

Encoder j

$$C_i = \sum_{j=1}^4 \alpha_{ij} h_j$$

aligner scores



$$C_1 = \alpha_{11} h_1 + \alpha_{12} h_2 + \alpha_{13} h_3 + \alpha_{14} h_4$$

Bahdanau Attention

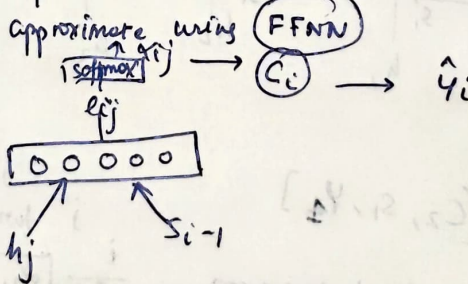
$$\alpha_{11} = f(h_1, s_0)$$

$$\alpha_{21} = f(h_1, s_1)$$

encoder current hidden state

decoder previous hidden state

$$\alpha_{ij} = f(h_j, s_{i-1})$$



1) Encode sentence

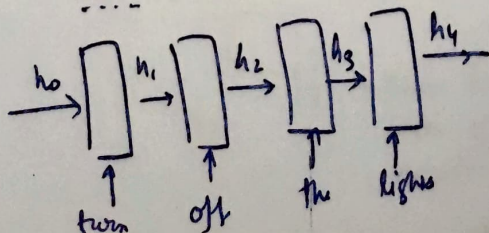
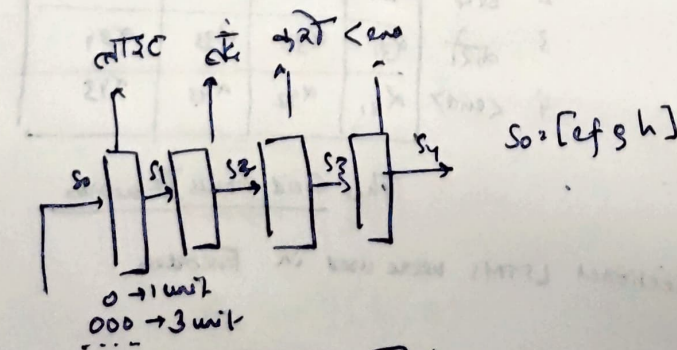
→ h_1, h_2, h_3, h_4

$i=1$

$$C_1 = \alpha_{11} h_1 + \alpha_{12} h_2 + \alpha_{13} h_3 + \alpha_{14} h_4$$

$$\alpha = f(h_j, s_{i-1})$$

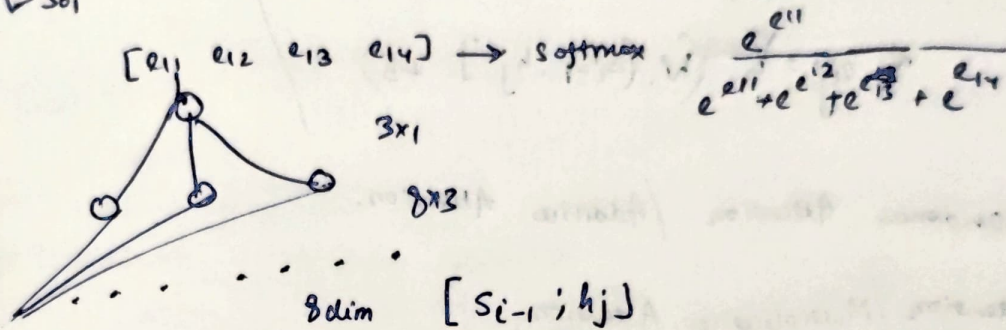
$$\alpha_{ij} = f(h_j, s_0)$$



$h_0 = [a \ b \ c \ d]$
4aim

4 rows x 8 cols.

	s_{01}	s_{02}	s_{03}	s_{04}	h_{11}	h_{12}	h_{13}	h_{14}
s_{01}								
s_{01}	s_{02}	s_{03}	s_{04}	h_{21}	h_{22}	h_{23}	h_{24}	
s_{01}	s_{02}	s_{03}	s_{04}	h_{31}	h_{32}	h_{33}	h_{34}	
s_{01}	s_{02}	s_{03}	s_{04}	h_{41}	h_{42}	h_{43}	h_{44}	

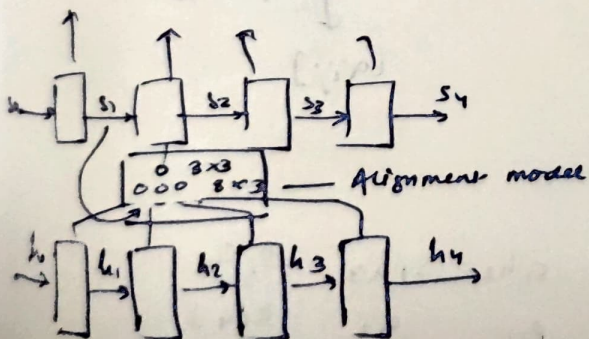


batch operation: $(4 \times 8) \times (8 \times 3) = (4 \times 3)$
 $\text{tanh}(4 \times 3) \times 1 \rightarrow (4 \times 3) \times (3 \times 1)$

$y_t \rightarrow (S_i)$
 $\uparrow \text{LSTM}$
 $[s_0, y_{t-1}, c_t]$

$t=2$
 $c_2 = \sum_{j=1}^{j=4} \alpha_{2j} h_j = \alpha_{21} h_1 + \alpha_{22} h_2 + \alpha_{23} h_3 + \alpha_{24} h_4$

s_p, c_2



s_{11}	s_{12}	s_{13}	s_{14}	h_{11}	h_{12}	h_{13}	h_{14}
s_{11}	s_{12}	s_{13}	s_{14}	h_{21}	h_{22}	h_{23}	h_{24}
s_{11}	s_{12}	s_{13}	s_{14}	h_{31}	h_{32}	h_{33}	h_{34}
s_{11}	s_{12}	s_{13}	s_{14}	h_{41}	h_{42}	h_{43}	h_{44}

4x8
time distributed FCNN

(s_i, h)

$(4 \times 8) \times (8 \times 3)$
 $(4 \times 3) \times (3 \times 1) \Rightarrow (4 \times 1)$

$e_{21} \ e_{22} \ e_{23} \ e_{24}$

$c_2, s_1, y_1 \rightarrow \text{LSTM} \rightarrow y_2$
 $\downarrow$
 vec/s_2

$$c_{ij} = \sum \alpha_{ij} h_j$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^4 \exp(e_{ik})}$$

softmax(e_{ij})

$$e_{ij} = \tanh(W \cdot [s_{i-1} : h_j] + b)$$

Bandwidth Attention / Additive Attention.

Luong Attention / Multiplicative Attention

$$c_i = \sum \alpha_{ij} h_j$$

$$\text{Bandwidth } \alpha_{ij} = f(s_{i-1}, h_j)$$

$$\text{Luong } \alpha_{ij} = f(s_i, h_j)$$

① Current (updated information)

② $s_i^T \cdot h_j \rightarrow$ dot product

$$s_i = [a \ b \ c \ d]$$

$$h_j = [e \ f \ g \ h]$$

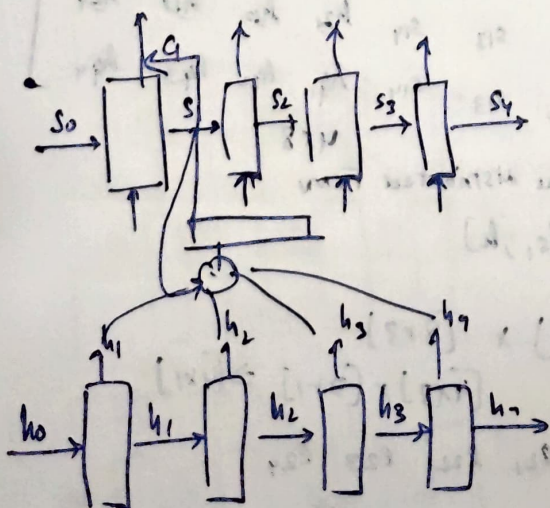
$$[ae + bf + cg + dh]$$

$$\downarrow \text{score}$$

$$[e_{ij}]$$

softmax

$$[\alpha_{ij}]$$



$$\begin{matrix} s_1 h_1 & s_1 h_2 & s_1 h_3 & s_1 h_4 \\ [e_{11} & e_{12} & e_{13} & e_{14}] \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \alpha_{11} & \alpha_{12} & \alpha_{13} & \alpha_{14} \end{matrix} \rightarrow [c_1]$$

$$[S_i c_i] \Rightarrow [y] \rightarrow \text{FFNN} + \text{softmax}$$